



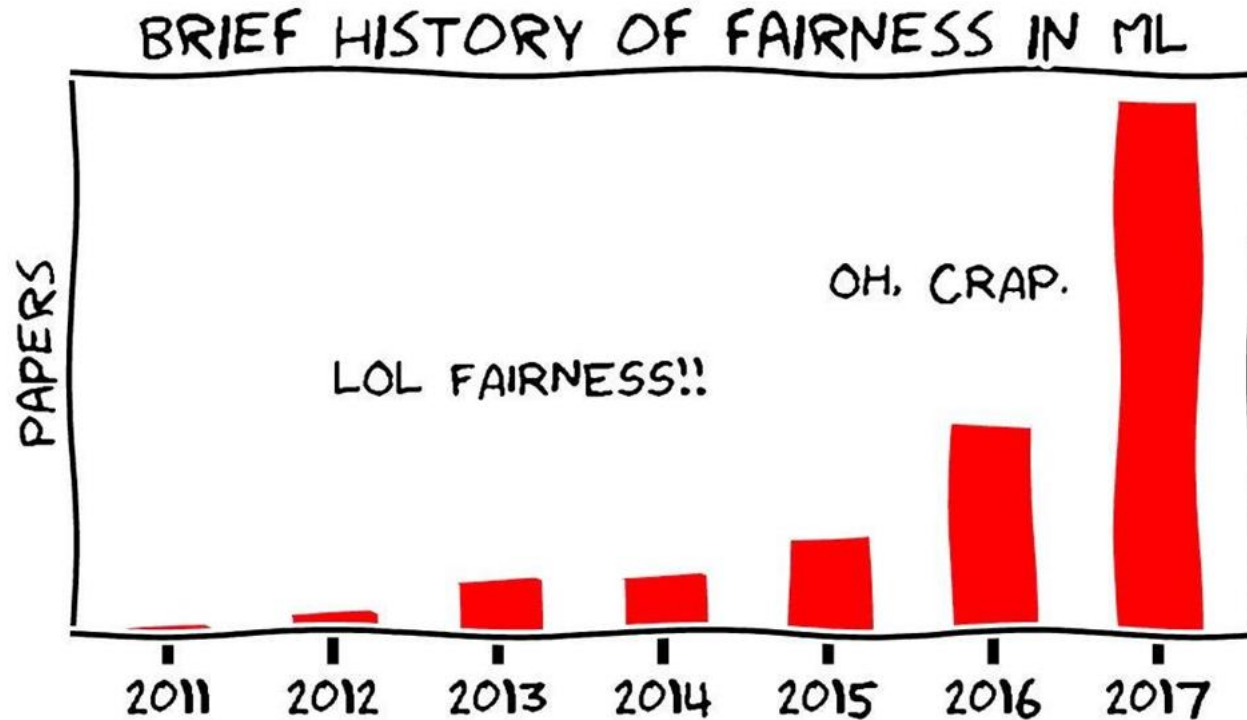
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EXPLAINABILITY FOR FAIR MACHINE LEARNING

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Fair Machine Learning



Fair Machine Learning - Challenges

- **Defining fairness** is hard:
 - Many competing definitions, often incompatible with each other
 - statistics-based, causal-reasoning-based, etc.
 - focus on group outcomes vs. individual outcomes
 - Requires contextual understanding

Name	Closest relative	Note
Statistical parity	Independence	Equivalent
Group fairness	Independence	Equivalent
Demographic parity	Independence	Equivalent
Conditional statistical parity	Independence	Relaxation
Equal opportunity	Separation	Relaxation
Equalized odds	Separation	Equivalent
Conditional procedure accuracy equality	Separation	Equivalent
Disparate mistreatment	Separation	Equivalent
Balance for positive class	Separation	Relaxation
Balance for negative class	Separation	Relaxation
Predictive equality	Separation	Relaxation
Conditional use accuracy equality	Sufficiency	Equivalence
Predictive parity	Sufficiency	Relaxation
Calibration	Sufficiency	Equivalence

Fairness definitions - Example

- **demographic parity: “ $f(x)$ be unconditionally independent of sensitive attr. a .”**

if 100 female students and 100 male students apply to Harvard University, **demographic parity** is achieved if **the percentage of** female students admitted is **the same** as **the percentage of** male students admitted, **irrespective of whether one group is on average more qualified than the other.**



Fairness definitions - Example

- **equalized odds: “ $f(x)$ be independent of sensitive attr. a given y ”**

if 100 female students and 100 male students apply to Harvard University, equalized odds is achieved if **qualified** female and male students both have **the same chance** of being **admitted**, and **unqualified** female and male students have **the same chance** of being **rejected**.

Female students

	Qualified	Unqualified
Admitted	45	2
Rejected	45	8
Total	90	10



Male students

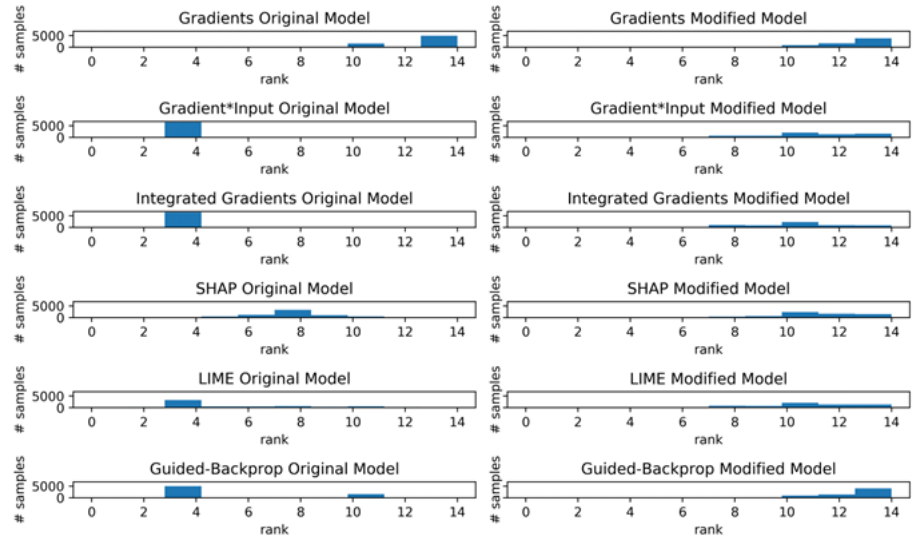
	Qualified	Unqualified
Admitted	5	18
Rejected	5	72
Total	10	90



Fair Machine Learning - Challenges

- Explanation methods can be **manipulated**
 - *[You shouldn't trust me: Learning models which conceal unfairness from multiple explanation methods, Dimanov, ECAI, 2020]*

Demonstrates that an explanation attack can easily mask a model's discriminatory use of a sensitive feature without hurting accuracy [Dimanov, 2020]



Importance ranking histograms for gender as the sensitive feature on the adult test set of the original (left) and modified (right) models.



Fair Machine Learning - Proposed Solution

- A **unified** approach that works for many **group-fairness** criteria
 - demographic parity, equalised odds, conditional demographic parity
 - for each definition, choose Shapley value functions that attribute overall fairness to individual features.
- **Cannot hide unfairness** by manipulating explanations
 - Fairness Shapley values collectively must sum to the chosen fairness metric





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Methodology

Explaining Model Accuracy

- If the team earns a total value $v(N)$, the Shapley value $\phi_v(i)$ attributes a portion to player i according to:

$$\phi_v(i) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)] \quad (1)$$

- Binary classification problem:

$$f_y(x) = (1 - y)(1 - f(x)) + y f(x) \quad (2)$$

- Define a value function by marginalising over out-of-coalition features:

$$v_{f_y(x)}(S) = \mathbb{E}_{p(x')} [f_y(x_S \sqcup x'_{N \setminus S})] \quad (3)$$

- Global explanation of the model's performance:

$$\Phi_f(i) = \mathbb{E}_{p(x,y)} [\phi_{f_y(x)}(i)] \quad (4)$$

- Aggregating global Shapley values

$$\sum_i \Phi_f(i) = \mathbb{E}_{p(x,y)} [f_y(x)] - \mathbb{E}_{p(x')p(y)} [f_y(x')] \quad (5)$$

Expected accuracy for a model which samples a predicted label according to the predicted probability

The accuracy that is not attributable to any of the features and is related to the class balance

Methodology

- Explainable Fairness:
 - How Shapley value paradigm can be adapted to explain fairness.
- Meta Algorithm:
 - Motivation: axiomatic properties of Shapley values
 - Applying existing training-time fairness interventions, wherein one trains a perturbation to the original model, rather than a new model entirely.



Explaining Model Fairness

- To explain fairness in a model's decisions, they define a new value function that captures this effect

Demographic parity calls for $f(x)$ to be unconditionally independent of a

$$g_a(x) = f(x) \cdot \frac{(-1)^a}{p(a)} \quad a: \text{sensitive attribute} \quad (6)$$

- The value function on coalitions is defined through marginalisation:

$$v_{g_a(x)}(S) = \mathbb{E}_{p(x')} [g_a(x_S \sqcup x'_{N \setminus S})] \quad (7)$$

$$\Phi_g(i) = \mathbb{E}_{\underline{p(x,a)}} [\phi_{g_a(x)}(i)] \quad (8)$$

The joint distribution of features and protected attribute from which the data is sampled

$$\sum_i \underline{\Phi_g(i)} = \int dx p(x|a=0) f(x) - \int dx p(x|a=1) f(x) \quad (9)$$

Each feature's marginal contribution to the overall demographic disparity in the model

Learning Corrective Perturbations

- The linearity axiom of the Shapley values guarantees that the fairness Shapley values of a linear ensemble of models are the corresponding linear combination of Shapley values of the underlying models.
- Motivated by this they consider the problem of learning an additive perturbation to an existing model in order to impose fairness.

$$f_{\theta} = f + \delta_{\theta} \tag{10}$$

$$\delta_{\theta}(f(x), x, a) = \sigma\left(\sigma^{-1}(f(x)) + \tilde{\delta}_{\theta}(f(x), x, a)\right) - f(x). \tag{11}$$

Any training-time fairness algorithm
used to learn the auxiliary
perturbation
e.g. Agarwal et al. (2018) and Zhang
et al. (2018)

The key idea is to reduce fair
classification to a sequence of cost-
sensitive classification problems,
whose solutions yield a randomized
classifier with the lowest (empirical)
error subject to the desired
constraints.



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Experiments & Results

Datasets

1. Adult dataset - UCI Machine Learning Repository (Dua & Graff, 2017)

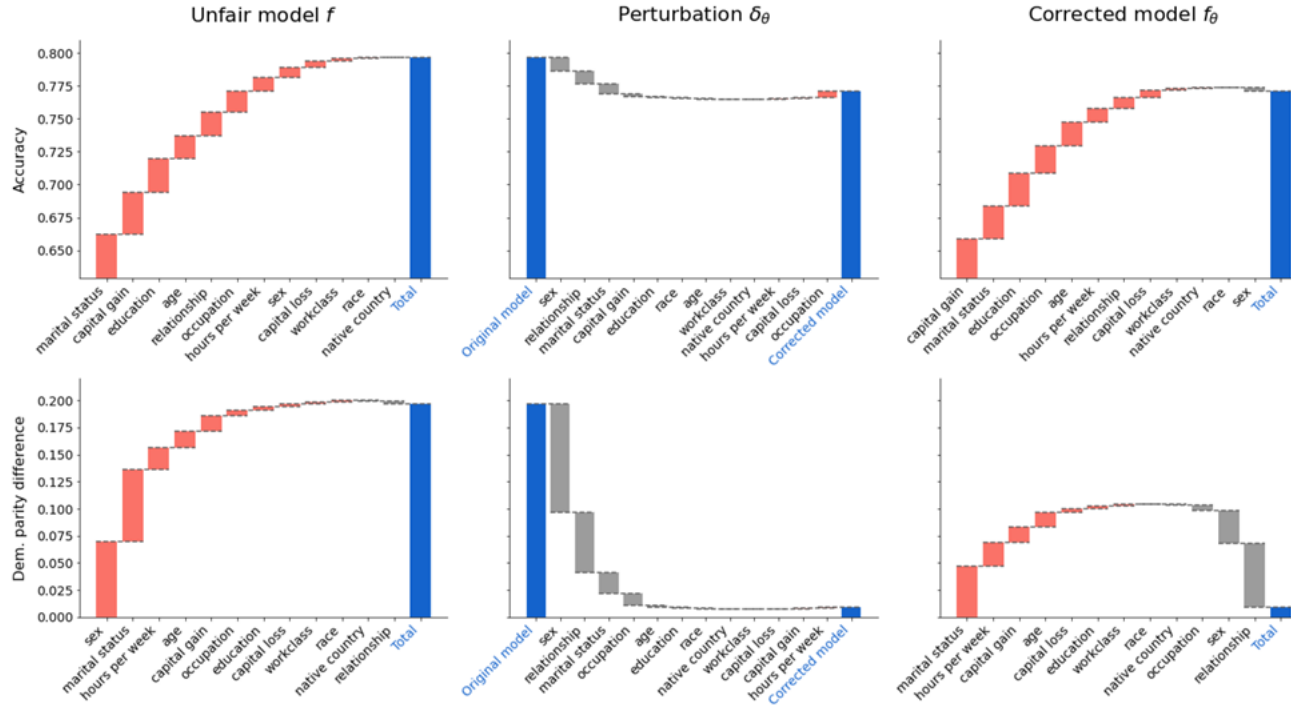
Task: predict whether an individual earns more than \$50K per year based on their demographics

1. COMPAS recidivism dataset (Larson et al., 2016)

Task: predict recidivism risk based on demographics



Explainability



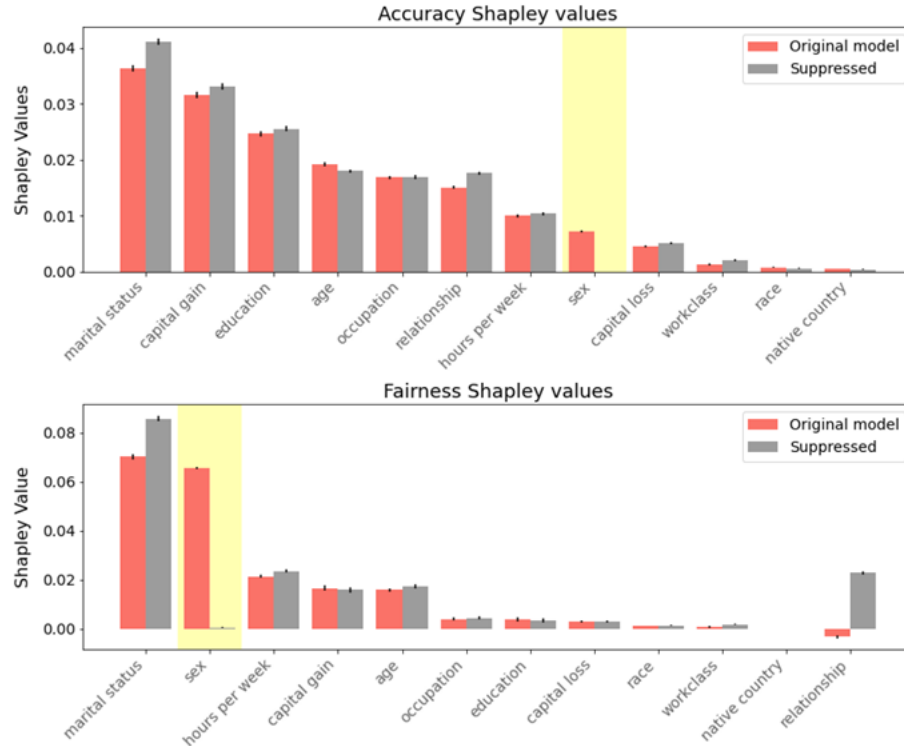
- Marital Status

- Sex

- Relationship



Robustness of Fairness Explanation



- Suppressing the importance of the sex feature.

- Demographic Parity Difference: 0.193 -> 0.184



Learnt Perturbations - Performance

Table 1: Accuracy associated with decreasing demographic parity thresholds.

		Accuracy [%] at demographic parity difference						
		0.1	0.08	0.06	0.04	0.02	0.01	0.005
Adult	Agarwal et al.	84.71	84.32	83.94	83.82	83.29	83.29	-
	Agarwal et al. - perturbed	84.69	84.43	83.82	83.82	83.35	83.23	-
	Zhang et al.	84.65	84.18	84.06	83.58	83.18	83.15	83.15
	Zhang et al. - perturbed	84.74	84.48	83.78	83.61	83.14	82.99	82.96
	Feldman et al. (post)	84.69	84.35	84.12	83.67	83.32	83.30	83.01
COMPAS	Agarwal et al.	74.05	74.05	73.77	73.67	73.11	73.11	73.01
	Agarwal et al. - perturbed	74.24	74.24	73.86	73.86	73.20	72.73	72.73
	Zhang et al.	75.19	75.19	75.19	74.62	74.15	74.15	74.15
	Zhang et al. - perturbed	74.24	74.24	74.24	73.30	73.30	73.20	72.73
	Feldman et al. (post)	74.81	74.81	74.81	74.24	74.24	73.20	72.35

- No significant reduction under the fairness definition of demographic parity.



Learnt Perturbations - Performance

Table 2: Accuracy associated with decreasing equalised odds thresholds.

	Method	Accuracy [%] at equalised odds difference						
		0.1	0.08	0.06	0.04	0.02	0.01	0.005
Adult	Agarwal et al.	85.32	85.32	85.13	84.30	84.18	-	-
	Agarwal et al. - perturbed	85.43	85.43	85.31	84.34	84.21	-	-
	Zhang et al.	85.13	85.04	85.04	84.86	84.33	75.43	75.43
	Zhang et al. - perturbed	85.26	85.11	85.06	84.97	84.23	83.53	-
	Hardt et al.	82.77	82.77	82.77	82.77	82.77	82.77	82.77
COMPAS	Agarwal et al.	75.19	75.19	74.05	74.05	73.86	73.39	73.39
	Agarwal et al. - perturbed	74.43	74.43	74.43	73.86	73.39	73.39	73.39
	Zhang et al.	74.62	74.62	74.62	74.62	74.62	73.48	53.12
	Zhang et al. - perturbed	74.34	74.34	74.34	74.34	73.48	72.44	72.44
	Hardt et al.	71.31	71.31	71.31	70.45	68.75	-	-

- No significant reduction under the fairness definition of equalised odds.

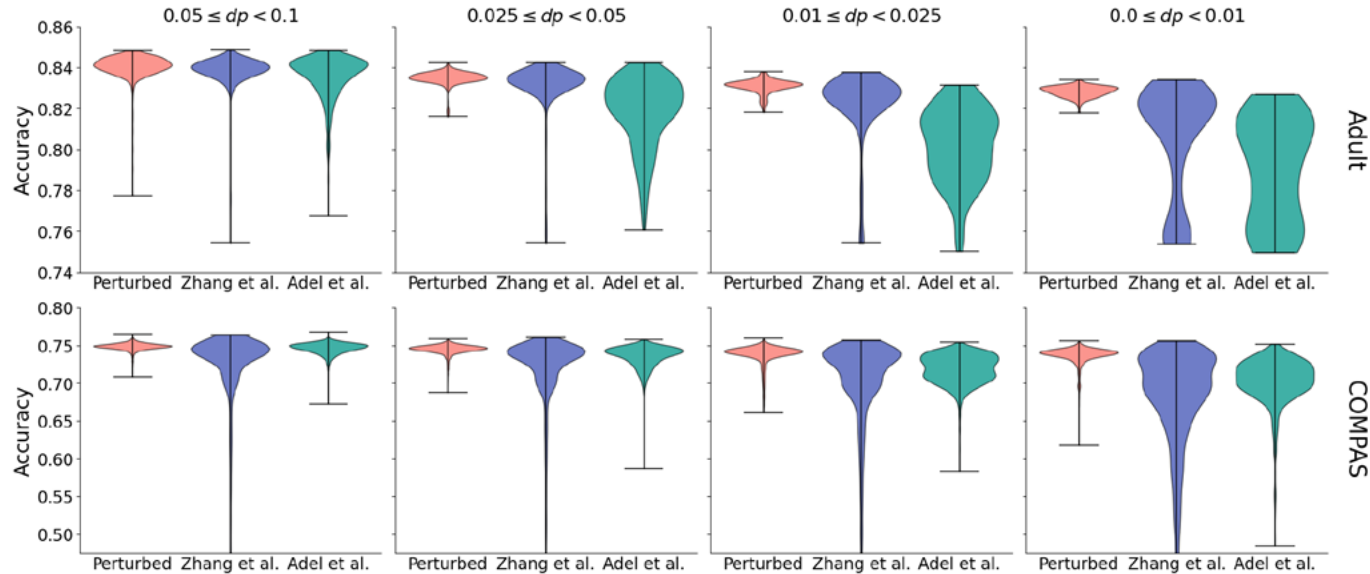


Learnt Perturbations - Flexibility

- Fully **model-agnostic** with respect to the original model, as any model structure or access requirements **apply only to the perturbation**, and not the original model.
- If the original model is complex, we have the option of **training a lightweight perturbation** to the complex model, and may not need to rerun an expensive training procedure.



Learnt Perturbations - Stability



- The proposed perturbative approach has less variance and higher mean accuracy.

Figure 3: Accuracy violin plots of experimental outcomes binned by achieved level of fairness.



Limitations & Discussion question

- What do you think are the advantages and disadvantages of the perturbation method proposed in the paper against other training-time fairness algorithm?
- From the fairness shapley value, if we observe that a certain feature contributes a lot to the unfairness (e.g., marital status in demographic parity difference), is it always correct to remove that particular feature from the model?
- After reading this paper, what's your opinion about using interpretability methods to validate the fairness of machine learning models?





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Thanks & Questions

Towards Bridging the Gaps between the Right to Explanation and the Right to be Forgotten

Satyapriya Krishna, Jiaqi Ma, Himabindu Lakkaraju

Motivation/Previous Work

- Right to an explanation: Given a model prediction, how can we provide a reason for this prediction to a user? (Algorithmic Recourse)
 - Multi-Objective Counterfactual (Dandel et al 2020)
 - MACE (model-agnostic counterfactual explanations) (Karimi et al 2020)
- Right to be forgotten: User can ask to have personal data removed from databases and models
 - Descent-to-delete (Neel et al 2021)
 - Approximate data deletion (Izzo et al 2021)

Motivation/Previous Work

- Pawelczyk et al. 2022: tradeoff between right to explanation and to be forgotten
 - Forgetting $\Rightarrow$ Model Changing $\Rightarrow$ Explanation not valid $\Rightarrow$ Right to explanation not met
- Trade-off stems from fact current explanation methods do not take into account underlying model changing
 - But! Wait! We know there are methods that do this like ROAR by Upadhyay et al. 2021
 - Authors say these methods assume certain model changes, but we can't know how a model may change as a result of forgetting
- Pawelczyk et al. highlighted the problem there is a need for a solution!

Contributions

- First algorithmic framework (ROCERF) to address tradeoff between right to explanation and to be forgotten
- ROCERF not only is able to bridge the gap but also outperform existing counterfactual explanation methods

ROCERF Framework

- **RO**bust **C**ounterfactual **E**xplanations under the **R**ight to be **F**orgotten
 - CFE as an optimization problem
 - CFE as an optimization problem under data removal
 - This is impractical! Efficient approximation
 - Theoretical bound on cost, and validity

ROCERF Framework: Notation

Training Data $D = \{(x_i, y_i)\}_{i=1}^n, x_i \in X, y_i \in \{-1, +1\}$

Data weight vector $\mathbf{w} \in \{0, 1\}^n$

Classifier Family $f_\theta : X \rightarrow \mathbb{R}, \theta \in \Theta$

$$y = \begin{cases} +1 & f_\theta(x) \geq 0 \\ -1 & f_\theta(x) < 0 \end{cases} \quad w_i = \begin{cases} 1 & \text{data point in training data} \\ 0 & \text{data point not in training data} \end{cases}$$

$\mathbf{w} = \mathbf{1} \Rightarrow$ no data point is being removed

ROCERF Framework: Notation

f_{θ_1} - trained on all data D , f_{θ_w} - trained on D_w

$$\theta_1 = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n l_i(\theta)$$

$$\theta_w = \arg \min_{\theta \in \Theta} \frac{1}{\|w\|_1} \sum_{i=1}^n w_i l_i(\theta)$$

$$g_i(\theta) := \frac{\partial l_i(\theta)}{\partial \theta}$$

$$h_i(\theta) := \frac{\partial g_i(\theta)}{\partial \theta^T}$$

$$H := \frac{1}{n} \sum_{i=1}^n h_i(\theta_1)$$

Counterfactual Explanation

- Optimisation problem - find valid counterfactual with minimum cost

$$\begin{aligned} & \min_{x \in \mathcal{X}} \quad \|x - x_0\|_2 \\ & \text{subject to} \quad f_{\hat{\theta}_1}(x) \geq 0. \end{aligned}$$

K-Removal Robust CFE

- k-RR CFE := CFE that is valid upon removal of any k data points

$$\mathcal{W}^{(k)} = \{w \in \{0, 1\}^n : \|w\|_1 = n - k\}$$

$$\min_{x \in \mathcal{X}} \|x - x_0\|_2$$

$$\text{subject to } f_{\hat{\theta}_w}(x) \geq 0, \forall w \in \mathcal{W}^{(k)}.$$

- Naive way - Train $\binom{n}{k}$ classifiers, and optimise with $\binom{n}{k}$ constraints
- Computationally impractical!

k-Robust Removal CFE: Efficient Approximation

- Note that classifier $f_{\hat{\theta}_w}$ is a function of both x and data weight vector $\mathbf{w}$
- Fix x and approximate $f_{\hat{\theta}_w}$ using Taylor series with respect to $\mathbf{w}$
- First order approximation!

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \frac{\partial f_{\hat{\theta}_w}(x)}{\partial w} \Big|_{w=1} (w - 1)$$

- Accounting for approximation error in the optimization problem

$$\begin{aligned} & \min_{x \in \mathcal{X}} \|x - x_0\|_2 \\ & \text{subject to } \tilde{f}_{\hat{\theta}_w}(x) \geq \delta, \forall w \in \mathcal{W}^{(k)}, \end{aligned}$$

k-RR CFE: Efficient Approximation

$$\begin{aligned}\tilde{f}_{\hat{\theta}_w}(x) &= f_{\hat{\theta}_1}(x) + \left. \frac{\partial f_{\hat{\theta}_w}(x)}{\partial w} \right|_{w=1} (w - 1) \\ &= f_{\hat{\theta}_1}(x) + \left. \frac{\partial f_{\theta}(x)}{\partial \theta} \right|_{\theta=\hat{\theta}_1} \left. \frac{\partial \hat{\theta}_w}{\partial w} \right|_{w=1} (w - 1),\end{aligned}$$

$$\left. \frac{\partial \hat{\theta}_w}{\partial w} \right|_{w=1} (w - 1) = \frac{1}{n} \sum_{i:w_i=0} H^{-1} g_i(\hat{\theta}_1). \quad \beta(x) := \left(\left. \frac{\partial f_{\theta}(x)}{\partial \theta} \right|_{\theta=\hat{\theta}_1} \right)^T$$

Giordano et al. (2019) “A swiss army infinitesimal jackknife”

The proof is left as an exercise to the reader!

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \frac{1}{n} \sum_{i:w_i=0} \beta(x)^T H^{-1} g_i(\hat{\theta}_1),$$

k-RR CFE: Efficient Approximation

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \frac{1}{n} \sum_{i:w_i=0} \beta(x)^T H^{-1} g_i(\hat{\theta}_1),$$

- We still have $\binom{n}{k}$ constraints
- The term $\beta(x)^T H^{-1} g_i(\hat{\theta}_1)$ is independent of data weight vector $\mathbf{w}$
- Pick the one tightest constraint!

k-RR CFE: Efficient Approximation

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \frac{1}{n} \sum_{i:w_i=0} \beta(x)^T H^{-1} g_i(\hat{\theta}_1),$$

$$\mathcal{A}(x) := \{\beta(x)^T H^{-1} g_i(\hat{\theta}_1)\}$$

$$f_{\mathcal{A}}^{(k)}(x) := f_{\hat{\theta}_1}(x) + \frac{1}{n} \min_{\mathcal{B} \subseteq \mathcal{A}(x), |\mathcal{B}|=k} \sum_{b \in \mathcal{B}} b,$$

$$\begin{aligned} & \min_{x \in \mathcal{X}} \quad \|x - x_0\|_2 \\ & \text{subject to} \quad f_{\mathcal{A}}^{(k)}(x) \geq \delta. \end{aligned}$$

k-RR CFE: Efficient Approximation

- One last step!
- Solving the constrained optimization problem.
- Penalty method

$$\phi(z) := \max(z, 0)^2$$

$$\min_{x \in \mathcal{X}} J_t(x) = \lambda_t \phi(\delta - f_{\mathcal{A}}^{(k)}(x)) + \|x - x_0\|_2,$$

k-RR CFE: Practical considerations

$$f_{\mathcal{A}}^{(k)}(x) := f_{\hat{\theta}_1}(x) + \frac{1}{n} \min_{\mathcal{B} \subseteq \mathcal{A}(x), |\mathcal{B}|=k} \sum_{b \in \mathcal{B}} b,$$

- Computational cost
 - $O(n)$
- Hyperparameters
 - k - number of data removals
 - δ - approximation error
- Linear Models
 - We can avoid backward pass!

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \frac{1}{n} \sum_{i:w_i=0} \beta(x)^T H^{-1} g_i(\hat{\theta}_1), \longrightarrow \tilde{f}_{\hat{\theta}_w}(x) = \hat{\theta}_1^T x + \frac{1}{n} \sum_{i:w_i=0} x^T H^{-1} g_i(\hat{\theta}_1),$$

Theoretical Analysis of Validity and Cost

Validity - Accuracy of the counterfactual explanation with respect to the retrained model.

$$\frac{1}{M} \sum_{w \in \mathcal{V}} \frac{1}{|\mathcal{T}|} \sum_{x \in \mathcal{T}} \mathbb{1}[f_{\hat{\theta}_w}(c(x)) = 1],$$

Cost - Distance measure for how far the the new input is from the original to satisfy the counterfactual requirement.

$$\frac{1}{|\mathcal{T}|} \sum_{x \in \mathcal{T}} \|c(x) - x\|_2.$$

Analysis on Linear Models

For regularized logistic regression model with loss function defined:

$$l_i(\theta) = \log(1 + \exp(-y_i \theta^T x_i)) + \gamma \|\theta\|_2^2.$$

The result states that the additional cost needed to achieve robust validity has an upper bound of $O(k/n)$ with theoretical guarantees for validity.

$$\|\tilde{x}_0^{(k)} - x_0\|_2 \leq \|\tilde{x}_0 - x_0\|_2 + \frac{kC}{n\|\hat{\theta}_1\|_2},$$

Assumptions for Nonlinear models

$$\sup_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \|h_i(\theta)\|_F \leq C_1;$$

$$\sup_{\theta \in \Theta} \|g_i(\theta)\|_2 \leq C_2, i = 1, \dots, n;$$

$$\sup_{x \in \mathcal{X}} \|\beta(x)\|_2 \leq C_3;$$

$$H(\theta) := \frac{1}{n} \sum_{i=1}^n h_i(\theta) \text{ is nonsingular and}$$

$$\sup_{\theta \in \Theta} \|H(\theta)^{-1}\|_{\text{op}} \leq C_4;$$

there exists suitable $\Delta > 0$, such that

$$\sup_{\|\theta - \hat{\theta}_1\|_2 < \Delta} \frac{1}{n} \sum_{i=1}^n \|h_i(\theta) - h_i(\hat{\theta}_1)\|_F \leq C_5 \|\theta - \hat{\theta}_1\|_2.$$

Analysis on Nonlinear Models

With certain regularity assumptions, the result provides theoretical guarantees where C is a constant independent of n .

$$\begin{aligned} & \|\tilde{x}_0^{(k)} - x_0\|_2 \\ & \leq \|\tilde{x}_0 - x_0\|_2 + \min_{\substack{x \in \mathcal{X}, \\ f_{\hat{\theta}_1}(x) - f_{\hat{\theta}_1}(\tilde{x}_0) \geq \frac{kC}{n}}} \|x - \tilde{x}_0\|_2, \end{aligned}$$

If the function is μ -strongly convex then,

$$\|\tilde{x}_0^{(k)} - x_0\|_2 \leq \|\tilde{x}_0 - x_0\|_2 + \frac{2kC}{n\mu}.$$

Experimental Setup

- Datasets - Three real-world binary classification datasets collected from high-stakes decision making scenarios.
- Predictive Models - Regularized logistic regression and 3-layer fully-connected feedforward neural network.
- Baseline Methods for Comparison - SCFE, C-CHVAE, and ROAR

Experimental Setup

- Validity is evaluated by randomly removing a small fraction (α) of the training data points (varies between 0.5% to 5%).
- Process is repeated M times (fixed to 100).
- Hyperparameters - k as 0.5% of the training set size and fix $\delta = 0$

Experimental Results

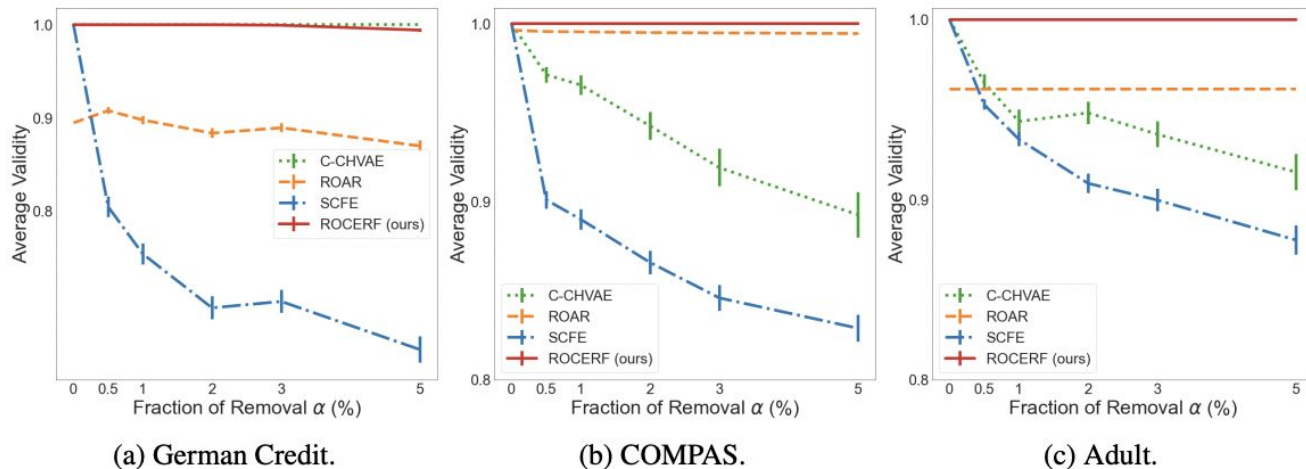
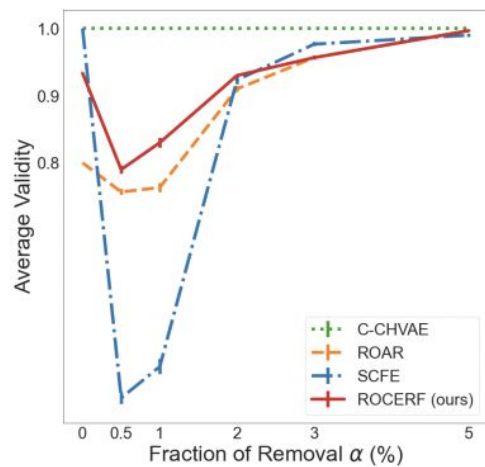
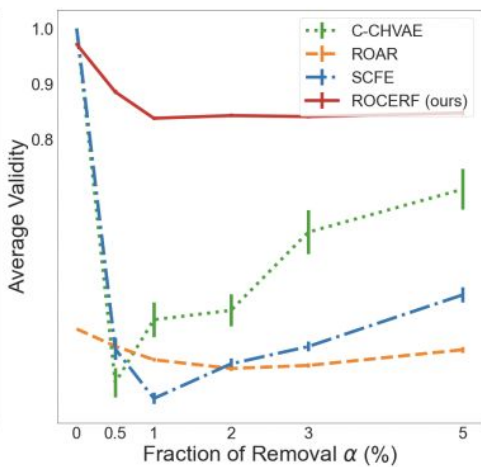


Figure 1. Average validity of different counterfactual explanation methods applied to logistic regression models on three datasets. In each figure, the x-axis corresponds to the fraction of data removal α and the y-axis corresponds to the average validity. The error bars indicate the standard errors across $M = 100$ trials with each trial having an α fraction of training data points randomly removed.

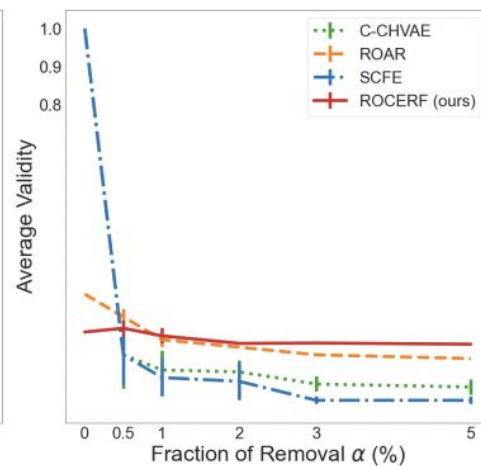
Experimental Results



(a) German Credit.



(b) COMPAS.



(c) Adult.

Figure 2. Average validity of different counterfactual explanation methods applied to neural network models on three datasets. See Figure 1 for more details about the plot setting.

Experimental Results

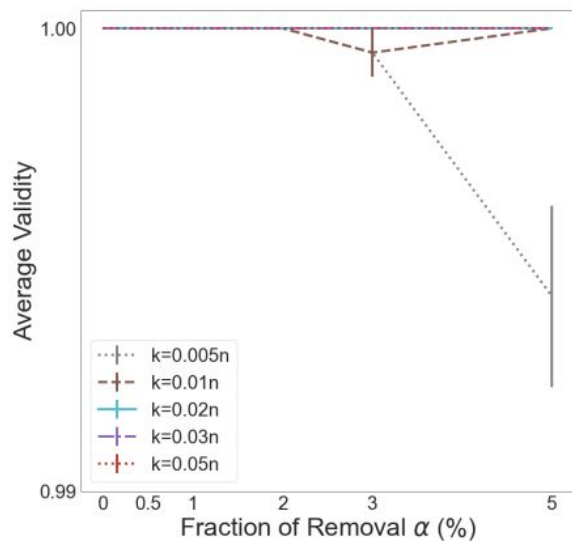
Methods	German Credit	COMPAS	Adult
SCFE	0.82 ± 0.12	0.78 ± 0.02	1.04 ± 0.006
C-CHVAE	8.51 ± 0.38	5.93 ± 0.11	3.79 ± 0.013
ROAR	1.45 ± 0.09	1.08 ± 0.01	1.07 ± 0.006
ROCERF (ours)	1.35 ± 0.14	0.87 ± 0.02	1.14 ± 0.006

Table 1. Average cost of different recourse methods applied to logistic regression models on three datasets. The cost is measured in terms of L2 norm.

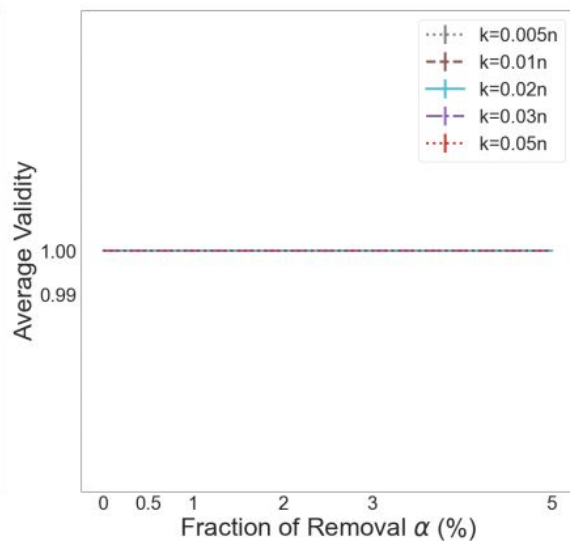
Methods	German Credit	COMPAS	Adult
SCFE	1.18 ± 0.08	0.97 ± 0.11	1.00 ± 0.09
C-CHVAE	4.45 ± 0.18	5.98 ± 0.12	8.83 ± 0.31
ROAR	3.84 ± 0.33	1.09 ± 0.13	4.07 ± 0.55
ROCERF (ours)	2.76 ± 0.22	3.07 ± 0.08	4.06 ± 0.52

Table 2. Average cost of different recourse methods applied to neural network models on three datasets. The cost is measured in terms of L2 norm.

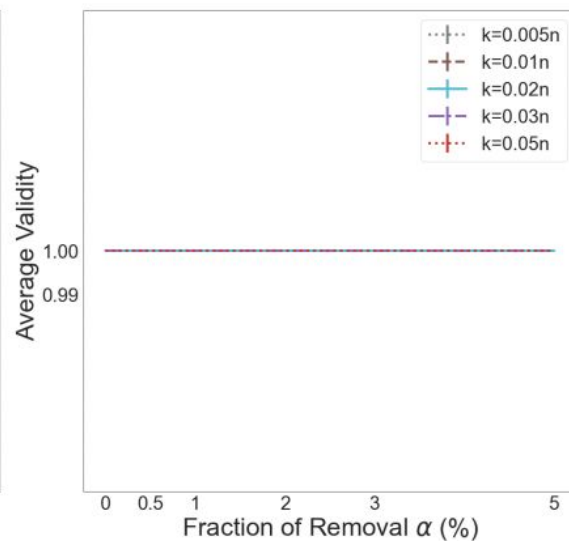
Experimental Results



(a) German Credit.



(b) COMPAS.



(c) Adult.

Figure 3. Sensitivity analysis with respect to the hyperparameter k .

Conclusions

- Propose ROCERF: Initial attempt to bridge the gap between right to be forgotten and right to explanation
- ROCERF is provably robust to model updates as triggered by data deletion requests
- ROCERF not only comes with theoretical guarantees on validity and cost but also outperforms baselines so is able to bridge this gap

Limitations/Future Work

- Non-linear models: Uses linear approximations under certain conditions, do these reflect non-linear model behavior? (common throughout recourse literature)
 - Some results show our assumptions may not hold for more complex non-linear neural networks (validity improves in the models as more data points are "forgotten")
- ROAR and ROCERF track closely for some of the results in this paper
 - Authors use ROAR with same hyperparameters as original paper, one could argue that ROAR could encapsulate ROCERF
- Why would the performance of ROCERF not vary more with k ? Test higher alphas?
 - Intuitively as more points are removed in the framework the model should be less robust but seems like it doesn't matter?

Discussion

- This is an example of when we were able to bridge the gap (as compared to privacy vs explanation where they seem more opposed). Any other tradeoffs that we learned about or that you think are able to be bridged?
- Given ROAR exists, do we need ROCERF? Could one not argue that the deletion of data parameters causes a perturbation of model parameters?
 - Why didn't the previous paper (Pawelczyk et al. 2022) use ROAR?
- Why has recourse literature focused on linear approximations when results show that non-linear models exhibit non-intuitive behavior?